REVENUE ANALYSIS

Learning Outcome

- evaluate a company's revenue and revenue drivers, including pricing power

Revenue Drivers

After determining the business model, we turn to the analysis of historical financial results and position for the company. This analysis will be used as a basis for forecasting a financial statement model that supports security valuation. For most companies—perhaps with the exception of banks, for which the balance sheet comes first—analysts start by analyzing revenues. As with other financial statement lines, the analysis involves identifying **drivers**, which are causative factors that explain the level of and changes in an output variable (here, revenues), and understanding the evolution of the drivers over time.

Analysts can take a bottom-up or top-down approach to determining revenue drivers. A bottom-up approach decomposes revenues into drivers such as sales volume and price, or revenues by product line, segment, or geography, which may be further broken down into other drivers. A top-down approach expresses revenues as a function of drivers such as market share, the addressable market or market size, and GDP growth. The two approaches are often used together.

Case Study: Bottom-Up Approach to Revenue Analysis: Warehouse Club Inc.

Warehouse's revenues are composed of net sales and membership fees.

Based on her business model work, Nguyen knows that net sales are sales of merchandise and services in Warehouse stores (the company does not operate an e-commerce business). A logical driver of net sales each year, therefore, is the number of stores open that year. Nguyen decomposes Warehouse's net sales into two drivers: the *average* number of stores open and net sales per average store. Nguyen also calculates the change in these drivers: the absolute change in stores each year (new-store openings; so far, Warehouse has never closed a store) and the annual percentage change in net sales per store.

	2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X
Stores, year-end	108	112	116	120	126	130	136	141	145	1
New-store openings	5	4	4	4	6	4	6	5	4	
Average stores (#)	106	110	114	118	123	128	133	139	143	1
Net sales per average store	138	150	162	166	170	169	166	173	184	1
Growth rate		9%	8%	2%	3%	−1%	−2%	4%	6%	5
Net sales	14,488	16,539	18,442	19,545	20,940	21,597	22,054	23,973	26,302	28,

An encouraging sign for forecasting this company's results is that each driver appears relatively stable, indicating that they might be somewhat predictable. Sometimes, however, key revenue drivers are volatile, such as interest rates and oil prices for banks and oil producers, respectively. Volatility does not mean that the analyst has selected an incorrect driver; drivers are based on the business model.

Nguyen then decomposes Warehouse's membership fee revenues into the annual price of membership and the average number of members.

Members	2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X19
Members, end of year	6,004	6,707	7,011	7,410	7,980	8,474	9,044	9,386	9,804	10,241
Members, annual average	5,900	6,360	6,855	7,218	7,691	8,218	8,764	9,217	9,600	10,017
Avg. per member fee	50.00	50.00	55.00	55.00	55.00	55.00	55.00	60.00	60.00	60.00
Membership fee revenues	295	318	377	397	423	452	482	553	576	601

Membership growth has been a larger driver of membership fee revenue growth than price increases, as the number of members has grown by 70% from 2X10 to 2X19 while prices have increased by 20% over the same period.

Management indicated that over 90% of members live within 10 kilometers of a store and, from 2X10 to 2X19, the company opened 46 new stores. Nguyen adds to her membership analysis by calculating the average number of members per store; if this number has been falling over time, it could mean that existing stores are losing members and sales to new stores the company opens, an example of **cannibalization**.

	2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X
Avg. members per store	57,282	57,818	60,128	61,171	62,528	64,205	65,892	66,546	67,133	68,
Growth rate		0.9%	4.0%	1.7%	2.2%	2.7%	2.6%	1.0%	0.9%	1.

Not only has the average number of members per store not declined, but it has also grown by 19%, indicating no cannibalization and that new-store openings are not the only driver of membership growth. This is a positive sign, though Nguyen notes that additional analysis is needed to estimate a ceiling on the number of members a store can reasonably support, and to determine whether stores that opened many years ago have already hit this ceiling.

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Pricing Power

Prices are a driver of revenue for every firm, and it may appear that pricing is a unilateral decision by management (e.g., Warehouse management chooses its markups on merchandise net sales and has chosen to increase its annual membership fee by $5 every five years). In fact, prices are constrained by a company's **pricing power**. Pricing power refers to a company's ability to set prices and other economic terms with customers without affecting its sales volumes. While Warehouse's management could choose to increase the price of membership from $60 to $600, such a decision would likely result in the loss of all its members. The membership price is constrained by the savings Warehouse can offer members on purchases, which itself is the subject of competition from other retailers that also use low prices to attract customers and of negotiations with suppliers that desire higher prices. Pricing power is primarily a function of both market

are price *takers*—that is, price is dictated by the forces of supply and demand—and all firms generally sell at the same price. If a firm prices above the prevailing market price, it will sell a fraction or none of its output, as customers will buy elsewhere. It is also generally not rational for a firm to price below the market price, as doing so would not only reduce its profits but also cause competitors to do the same, leading to a price war. Returns on capital in highly competitive markets move in cycles based on supply and demand, but over the long run approximate firms' cost of capital as competition forces prices down to marginal cost. A **low-cost producer** (with costs below—in some cases, *well* below—those of a marginal producer) can, however, earn long-run returns above its cost of capital in a highly competitive industry, but it requires the low-cost position to be sustained against competition over time. A prominent example is state-owned oil producers that benefit from monopoly ownership of low-cost oil reserves in certain countries.

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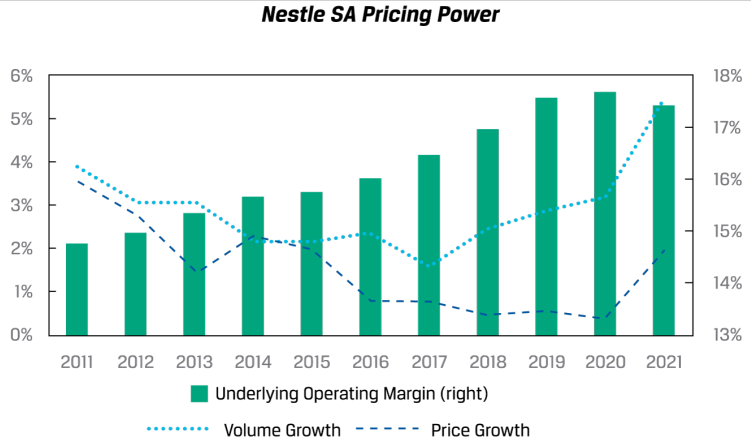
Other attributes of highly competitive markets include little to no product differentiation, low barriers to firm entry, available substitutes, a lack of customer loyalty, and low switching costs for customers. Many markets fit this description, including retail, oil and gas and other natural resources, bank loans and deposits, insurance, fresh food, off-patent pharmaceuticals, transportation, and some types of technology hardware. Highly competitive markets often do not start as highly competitive but, rather, become more competitive as new firms enter, the pace of innovation slows, and imitation becomes widespread, a process known as **commoditization**.

Firms operating in less competitive markets, such as monopolistic competition, oligopoly, or monopoly structures, tend to have some degree of pricing power. They can raise prices or change other economic terms (contract length, late fees, etc.) without materially affecting sales volumes. Attributes of these markets include product differentiation, barriers to entry, lack of economical substitutes, switching costs for customers, and a high degree of customer loyalty or retention. Markets and companies that fit this description include branded consumer goods, utilities, patented pharmaceuticals, software, restaurants, payment networks, medical devices, and some types of real estate. Firms with pricing power often apply value- or cost-based pricing as well as price discrimination strategies (outlined in the previous module). While product differentiation is most visible, differentiation can also include convenience, wrap-around services, and a secondary resale market.

Analysts evaluate pricing power not just by examining a firm's prices over time or relative to competitors, but also by comparing a firm's prices with its costs—in other words, its profit margins. While a company may be increasing prices by 3% per year, if its costs are rising by 5%, it is showing both an inability to pass along price increases to customers and declining profitability for investors. A company without pricing power may be unable to increase prices even as costs rise, because it is constrained by the availability of substitutes and customers can easily switch. For example, a movie theater might be unable to increase prices in line with costs, because streaming video subscriptions are cheaper and the company would likely lose customers if it raised prices significantly. Therefore, an important sign of pricing power is rising profitability over time, which demonstrates both an ability to retain economics for investors and that competitors, new entrants, or substitutes are not driving down prices faster than costs.

EXAMPLE 2

Nestlé SA is a global packaged food and beverage company headquartered in Switzerland. The company sells products with proprietary brand trademarks (such as Nespresso, Purina, Perrier, Maggi, and Stouffer's) that it acquires and develops with marketing and R&D investments, which totaled CHF19 billion in 2021, or 20% of the company's revenues. The distinctiveness of its products through branding can confer



Top-Down Revenue Analysis

Analysts can also take a top-down approach to analyzing revenue by expressing it as a function of drivers such as market size and a company's share of that market. This approach will be discussed in more detail in the next module (on industry and competitive analysis), but we will demonstrate this approach for Warehouse Club Inc.

Case Study: Top-Down Approach to Revenue Analysis: Warehouse Club Inc.

A government agency reports total national retail and food services sales each month for the country that Warehouse operates in (the United States). The data capture consumer spending at hundreds of thousands of retailers and restaurants. Sales are categorized by the type of retailer or restaurant, which reflects the primary type of product or service it sells. An excerpt of a monthly release is shown below.

National Retail Sales, January 2X19 (billions of USD)	
Total retail and food services sales	456,012
Total retail sales	399,359
Automobile and parts dealers	88,152
Petrol stations	36,539
Pharmacies and drug stores	24,273
Grocery stores	57,392
Total food services sales	56,653
Full-service restaurants	24,730
Limited-service restaurants	25,273
Pubs, bars, and other	6,650

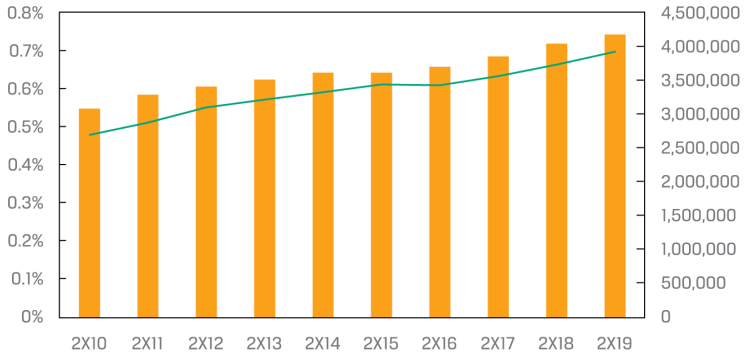
Warehouse sells a broad range of retail goods in most cities, so US national retail spending is generally representative of Warehouse's **market size**—the existing demand for goods or services offered by a company. By expressing Warehouse's revenues as a percentage of the market, or its **market share**, over time, we can assess whether the company has been gaining or losing relevance with its customers relative to competitors and substitutes.

A challenge for analysts is determining market size. The data might be difficult to gather (not the case here), and judgment is required for determining what to include and exclude: only sales of products identical to the company's, sales of similar products, all sales of competitors, and so on. The common practice is to include all sales of *similar* products and to consider *substitutes* separately (discussed in the next module on

market that a firm does not have (i.e., 100% – *x*% market share) represents sales potential.

Elaine Nguyen decides to use total US retail sales as the market size for Warehouse, with one exclusion: automobiles and parts. Warehouse does not sell automobiles or automobile parts, nor does it sell substitute means of transportation or repair. Nguyen considers food service substitutes (restaurants and pubs) for Warehouse.

Warehouse Club Inc. Market Share (%, left) of Retail Sales excl. Autos and Parts (USD millions, right)



Nguyen makes three observations:

- The market is large and growth has been positive, albeit at a slow (3.4%) rate. This finding aligns with Nguyen's expectations, because this is a developed market that represents a broad range of consumer spending (a large portion of which is for necessities) and the country has been in an economic expansion.
- Warehouse Club Inc. has a small (70 basis points in 2X19) but growing share of the market (up from 48 basis points in 2X10), indicating that its value proposition relative to other retailers has been resonating with consumers.
- Warehouse's market share increased in most years, with the exception of 2X16, when its share was flat from the prior year. Nguyen will have to investigate further.

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Discussion Board Question

Which of Nguyen's analyses was more helpful to you in understanding Warehouse's historical revenue growth: the bottom-up or the top-down approach? Why?

We encourage you to post your response, as well as read and reply to others' responses, on the Learning Ecosystem discussion board.

Question Set

Questions 1–4 refer to the earlier vignette on Iliso Marketplace Ltd.

1. In 2X19, the percentage of GMV that Iliso collected from third-party merchants (the company's "take rate") was *closest* to:

A. 2.2%.
B. 14.7%.
C. 15.0%.

Hide Solution

percentage of the value of transactions facilitated by the platform. In 2X19, Iliso third-party merchant fees were $190 million against third-party gross merchandise value (GMV) of $1,269 million on its platform. The take rate is therefore:

190/1,269 = 0.1497 ≈ 15.0%

A is incorrect, because it results from dividing third-party merchant fees by total GMV. B is incorrect, because 14.7% is the percentage of Iliso's GMV attributable to third-party merchant sales.

2. Which of the following measures would be used to best assess Iliso's pricing power?

A. Take rate
B. GMV year-on-year growth rate
C. GMV per active customer account

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Solution:

A is correct. The percentage of transaction value retained by the platform reflects the price Iliso charges third-party merchants to use its platform. Both the stability and the level of its take rate reflect the state of competition that Iliso faces from other e-commerce platforms.

B is incorrect, because the GMV year-on-year growth rate can be affected by numerous factors, including customer volumes, product mix, and basket size. It does not carry information for analysts to estimate pricing power.

C is incorrect, because GMV per active customer account can be affected by product mix and the average basket size, which is driven by its $25 minimum for free shipping. This measure does not carry information for analysts to estimate the pricing power of the platform.

3. Referring to the data table earlier in this lesson titled "National Retail Sales, January 2X19" and assuming 2X19 US monthly retail and food sales of $456 billion and assuming Iliso's addressable geography is the United States, Iliso's market share of retail and food sales was *closest* to:

A. 0.134%.
B. 0.138%.
C. 0.157%.

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Solution:

C is correct. In 2X19, annual US retail and food sales were: 456 × 12 = $5,472 billion. Since GMV reflects the value of retail sales on Iliso's platform, the appropriate estimate of Iliso's market share is therefore:

2X19 GMV/2X19 US retail and food sales = 8,605/5,472,000 = 0.157%

A is incorrect, because this measure uses only Iliso's retail sales and ignores third-party GMV, even though the vignette states that merchants may sell the same products.

B is incorrect, because this measure uses Iliso's net revenues as a percentage of the market. This measure is inappropriate, since Iliso's net revenues do not reflect the gross value of third-party merchant sales on its platform.

4. Which of the following is an appropriate bottom-up driver of Iliso's revenues?

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Solution:

B is correct. In a bottom-up approach, analysts decompose the revenues into logical revenue drivers such as product line or segment. A driver of Iliso's net sales must be the average number of active customer accounts during the period and the net retail sales per account. We use GMV, since this measure accurately reflects the total amount of retail sales (both Iliso's first-party and third-party sales).

A and C are incorrect, because they are top-down, not bottom-up, drivers.

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Discussion

What do you want to discuss... (required)

I would say Bottom-up, it links revenue directly to observable company drivers (stores, sales per store, members).
Top-down adds useful market context but requires broader assumptions about market size and substitutes.
Together they're ideal, but bottom-up best explains historical growth.

Created 3 months ago by Paul Marquez — 0 replies | Last Activity: 3 months ago

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DM

I believe taking a top down approach would be better to understand the historic revenue growth of Warehouse Inc , as it would allow an analyst to compare the growth relative to GDP , Inflation , impacts due to monetary and fiscal policy , and then moving into the individual sectors of the company . This would allow the analyst to understand the revenue growth in most definitive and accurate manner .

Created 3 months ago by DRISHTI MODI — 0 replies | Last Activity: 3 months ago

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BO

I think to get an understanding of growth it is nice to see from a top-bottom perspective to see the full picture. Its like looking at it in a micro and macro perspective except the macro is more micro, focusing on a certain part of the market via industry or region. If were to understand whether a company is worth investing and if it's trading at reason multiples then I'd like to get an understanding of the total market share, how much of the market share does it control, and its growth rate compared to its competitors. Then get an understanding if the stock is trading at reasonable multiples. The company might be doing amazingly from the outside looking in but until you look in you won't know if it's built on top of straws.

Created 4 months ago by Brian Oxmar Moreno — 0 replies | Last Activity: 4 months ago

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YA

Well, I think both approaches can be seen from a different angle. The bottom-up approach was more intrinsic, whereas the top-down approach adjusted for the external factor (market size/share). One cannot go without the other, because if, as an analyst, you base your revenue drivers solely on average store opened or average price per new member (in the case of Warehouse Inc.), you must first ascertain the addressable market, so that when you approach forecasting based on average store opened, you already know the possible scope or space that is left in the said market for expansion.

Created 5 months ago by Yash Anand — 0 replies | Last Activity: 5 months ago

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PC

Bottom up Approach was more helpful in understanding historical revenue because it was compared with growth in number of customers, average number of stores, customers per average store which area all taken from historical data that is available from Warehouse Inc. The top-down approach required more assumptions and estimations from public or proprietary data available to the researcher.

Created 5 months ago by Priyesh Choodamani Iyer — 0 replies | Last Activity: 5 months ago

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AK

Bottom up approach v Top Down approach. Bottom Up approach for Warehouse Club Inc. seems to provide more promising estimates and can be used for deriving forecast with more precision. This is mainly due to readily available data. Top Down approach on the other hand is not very precise but based on analysts' best judgement. However Top down approach gives a solid idea about the positioning of the firm and the potential to grow with respect to the competition. In that sense Top Down also becomes significant approach. Using both are ideal for company analysis.

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